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Mini-Project Report on

“Kannada Morphological Analyzer and Generator Using Deep Learning”

SUBMITTED TO : Dr. Azeez Musthafa
NAME OF THE STUDENT : Ayishath Farha, Ayishatul Nafla,
Ayshath Afna
REGISTER NUMBER : U05BN23S0009,U05BN23S0034,
U05BN23S0007

UNDER THE SUPERVISION OF

Dr. AZEEZ MUSTHAFA

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DISSERTATION

VI-SEMESTER B.C.A 2025-2026

Name of The Student : Ayishath Farha, Ayishatul Nafla,
Ayshath Afna

Register Number : U05BN23S0009, U05BN23S0034,
U05BN23S0007

Subject : MINI-PROJECT

Date Of Examination :23/05/2026

Certified that this is the bonafide record of work done by the
Ayishath Farha, Ayishatul Nafla, Ayshath Afna ,
during the year 2025-2026 in the Computer Science Laboratory as prescribed by the
Mangalore University.

Lecturer – in –charge

Principal

Examiner's Signature:

1. _____

2. _____

DECLARATION

We hereby declare that the project entitled “**Kannada Morphological Analyser and Deep Learning Model**” has been carried out by us during the academic year **2025–2026**, in the **6th semester** of the Bachelor of Computer Applications (BCA) program, under the valuable guidance and supervision of **Mr. Azeez Musthafa, Department of BCA, Bearys Institute of Emerging Sciences**, in partial fulfillment of the requirements for the award of the **Bachelor’s Degree in Computer Applications (BCA)**.

We further declare that this project is the result of my/our own efforts and has not been submitted to any other university or institution for the award of any degree or diploma.

Place: Inoli End,Boliyar

Date: 23-05-2026

Name of the students:

Ayishath Farha

Ayishatul Nafla

Ayshath Afna

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ABSTRACT

The project entitled “**Kannada Morphological Analyser and Deep Learning Model**” for the academic year **2025–2026** is designed to analyze and process Kannada words by identifying their morphological structure and linguistic patterns using advanced deep learning techniques. The system focuses on breaking down Kannada words into their root forms, prefixes, suffixes, and grammatical components to improve natural language understanding and processing.

The project utilizes deep learning models trained on Kannada linguistic datasets to perform accurate morphological analysis and word prediction. By applying machine learning and natural language processing techniques, the system can efficiently recognize word forms, classify grammatical categories, and support language-based applications such as text processing, spell checking, machine translation, and information retrieval.

A user-friendly interface is developed to allow users to input Kannada text and obtain detailed morphological analysis results. The project was carried out under the valuable guidance of **Mr. Azeez Musthafa, Department of BCA, Bearys Institute of Emerging Sciences**.

The primary aim of this project is to contribute to the development of Kannada language technology by improving computational understanding of the language and supporting

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“KANNADA MORPHOLOGICAL ANALYSER AND GENERATOR USING DEEP LEARNING”

CHAPTER-1 INTRODUCTION

The Kannada Morphological Analyzer and Generator is a Natural Language Processing (NLP) based system developed to analyze and generate word forms in the Kannada language using Deep Learning techniques. Kannada is a morphologically rich and agglutinative language, where words are formed by adding multiple suffixes to a root word. Because of this complexity, traditional rulebased systems are not always efficient.

This project aims to build an intelligent system that can:

- ☐ Analyze Kannada words into root and grammatical components
- ☐ Generate correct word forms based on tense, gender, number, and case
- ☐ Improve regional language processing using Artificial Intelligence

The system will help bridge the gap between regional languages and modern AI-based applications.

1.1 OBJECTIVES

Objectives of Kannada Morphological Analyser and Generator:

1. To design and implement a deep learning model(Seq2Seq with LSTMs) for morphological Correction in Kannada.
2. To build a character-level vocabulary for source (incorrect text) and target(correct text).
3. To integrate a POS tagging dictionary for linguistic annotation.
4. To design a GUI application in Tkinter for real-time correction and analysis.
5. To improve Kannada language processing in AI applications like translation, chatbots, and voice assistants.
6. To promote research and development in regional language technology.

1.2 SCOPE

1.2.1 Word Analysis:

The system will break Kannada words into:

- Root word
- Suffix
- Grammatical tags (tense, gender, number, case)

1.2.2 Word Generation:

The system will generate different word forms from a given root word based on grammatical system.

1.2.3 Integration with NLP Applications:

The model can be integrated into:

- Machine Translation Systems
- Spell Checkers
- Grammar Correction Tools
- Educational Applications

1.2.4 Research Expansion:

The project can be extended to other Indian languages in the future.

1.3 Methodology

1.3.1 Dataset

Collected dataset in CSV format (data.csv) containing pairs:

- A. Incorrect form
- B. Correct form

Example: Incorrect: "ಕಂಡಿೞಿ" Correct: "ಕಂಡಿದ್ದು ಲೇ' ' 'ಲೇ' ' 'ನೆ"

1.3.2 Preprocessing

- Character-level tokenization.
- Special tokens added: , , , .

- Builtsource vocabulary (SRC_vocab) and target vocabulary (TRG_vocab).
- Saved vocabularies in JSON for future use.

1.3.3 Model Architecture

Implemented Encoder-Decoder Seq2Seq model:

- **Encoder:**
 - Embedding layer (dim=128)
 - 2-layer LSTM (hidden=256, dropout=0.5)
- **Decoder:**
 - Embedding layer (dim=128)
 - 2-layer LSTM (hidden=256)
 - Fully connected layer projecting to target vocabulary size.
- **Seq2Seq wrapper:**
 - Teacher forcing during training.
 - Cross-entropy loss with padding ignored.
 - Optimizer: Adam.

1.3.4 Training

- **Epochs:** 30
- Loss function: CrossEntropyLoss (ignore)
- Modelsaved as kannada_morph_model.pt.
- Vocabulary saved as SRC_vocab.json and TRG_vocab.json.

1.3.5 Inference

Input text passed through encoder → decoder .generates corrected sequence.

1.4 STRUCTURE OF THE PROGRAM

This project contains the following modules:

1.4.1 Admin Module:

The admin manages the system and dataset. Admin can perform the following activities:

- Add training datasets
- Update linguistic data
- Monitor model performance
- Retrain deep learning model
- Manage user access

- View analysis logs

1.4.2 User Module:

User has a separate interface. User can perform the following activities:

- Login securely
- Enter Kannada word for analysis
- View root word and grammatical details
- Enter root word with grammatical features
- Generate correct Kannada word form
- View previous analysis history

1.4.3 Model Module:

This module includes:

- Data Preprocessing
- Tokenization
- Unicode Normalization
- Deep Learning Model (LSTM / BiLSTM / Transformer)
- Sequence-to-Sequence architecture

1.5 EXPECTED OUTCOMES

- A functional deep learning-based Kannada morphological analyser.
- A morphological generator capable of reconstructing valid Kannada words.
- A reusable annotated dataset for future research.
- Demonstrated improvement over traditional rule-based systems.

1.6 TOOLS AND TECHNOLOGIES

- Programming Language: Python
- Frameworks: TensorFlow / PyTorch
- Libraries: NLTK, Hugging Face Transformers, OpenNMT
- Dataset: Custom-annotated Kannada corpus
- Environment: Jupyter Notebook / Google Colab/ Anaconda

1.7 APPLICATIONS

- Machine Translation (Kannada ↔ English)
- Spell-checking and grammar correction tools
- Education tool for students learning Kannada grammar.

- Assistive technology for non-native Kannada learners.
- Language learning and educational application

CHAPTER-2

Software Requirements Specification (SRS)

2.1 Introduction

2.1.1 Purpose

The purpose of this project is to design and develop a deep learning–based Kannada morphological analyzer and generator. The system will analyze Kannada words into their root forms and grammatical components, and generate valid Kannada word forms based on linguistic features provided by the user. This will support Kannada Natural Language Processing (NLP) tasks and enhance applications such as machine translation, grammar checking, and educational tools.

2.1.2 Scope

The system will:

- Identify the internal structure of Kannada words.
- Generate grammatically correct word forms based on linguistic attributes.
- Provide support for applications including machine translation, spell checking, text-to-speech, and chatbots.
- Serve as a foundational tool for Kannada language research and NLP applications.

2.1.3 Objectives

- Develop a morphological analyser for Kannada words.
- Develop a morphological generator for producing valid inflections.
- Apply deep learning techniques to improve accuracy.
- Support Kannada language processing tasks in diverse applications.

2.1.4 Stakeholders

- Primary Users: Students, researchers, linguists, educators.
- Secondary Users: Developers integrating Kannada NLP into applications.
- Maintainers: System administrators and NLP engineers.

2.2 Overall Description

2.2.1 System Perspective

The system consists of two major components:

- Morphological Analyzer: Extracts root words and grammatical information.
- Morphological Generator: Produces valid inflected word forms.

2.2.2 System Functions

- Input processing and text normalization.
- Morphological analysis using linguistic rules and deep learning.
- Morphological generation based on user-specified attributes.
- Output display with clear results.

2.2.3 User Characteristics

Users are expected to have basic knowledge of Kannada language. Technical users (researchers, developers) may integrate the system into larger NLP pipelines.

2.2.4 Constraints

- Must support Kannada Unicode text.
- Requires reliable datasets for training.
- GPU recommended for efficient training.

2.2.5 Assumptions and Dependencies

- Availability of annotated Kannada corpora.
- Dependence on Python libraries and frameworks (TensorFlow/PyTorch, NLTK).
- Integration with databases (MySQL/MongoDB).

2.3. Functional Requirements

2.3.1 Morphological Analyser

- Accept Kannada words/text as input.
- Identify root word/stem.
- Detect suffixes and inflections.
- Extract grammatical attributes (gender, number, person, tense, case, part of speech).
- Display results clearly.

Example: Input: ಮನೆಗಳಲ್ಲಿ Output: Root: ಮನೆ, Suffix: ಗಳಲ್ಲಿ Number: Plural, Case: Locative, Meaning: "In the houses."

2.3.2 Morphological Generator

- Accept root word + attributes as input.
- Generate grammatically valid word forms.
- Support noun and verb inflections
- Handle tense, gender, person, number.

Example: Input: Root:ಮನೆ, Number: Plural, Case: Locative **Output:**
ಮನೆಗಳಲ್ಲಿ

2.3.3 Deep Learning Module

- Train on Kannada linguistic datasets.
- Predict unseen words with reasonable accuracy.
- Support retraining and dataset expansion.
- Improve accuracy through continuous learning.

2.4 Non-Functional Requirements

Usability: Simple, user-friendly interface.

Performance: Fast response times(<1 second for analysis).

Reliability: Consistent and accurate outputs.

Scalability: Handle large datasets and future enhancements.

Security: Ensure safe handling of inputs and stored data.

2.5 Hardware Requirements

- Processor: Intel Core i5 or above.
- RAM: Minimum 8 GB.
- Storage: Minimum 256 GB.
- GPU: Optional, recommended for faster training.

2.6 Software Requirements

- Operating System: Windows or Linux.
- Programming Language: Python.
- Libraries/Frameworks: TensorFlow/PyTorch, NumPy, Pandas, NLTK, Flask/Django.
- Database: MySQL or MongoDB.

2.7 System Architecture

Modules include

- Input Module
- Text Preprocessing Module
- Morphological Analysis Module
- Deep Learning Prediction Module
- Morphological Generation Module
- Output Display Module

2.8 Use Case Scenarios

Use Case 1: Morphological Analysis Input: ಹ ಗುತ್ರಿ ಧ್ವಂಸ Output: Root: ಹ ಗು, Tense: Present Continuous, Person: Third, Gender: Male, Number: Singular.

Use Case 2: Morphological Generation Input: Root: ಹ ಗು, Tense: Past, Person: Third, Gender: Male Output: ಹ ಧ್ವ

2.9 Applications

- Machine translation systems.
- Spell and grammar checking tools.
- Educational applications.
- Chatbots.
- Text-to-speech and speech-to-text systems.
- Kannada language research and NLP applications.

CHAPTER-3

DETAILED DESIGN

3.1 Architecture Overview

The proposed system is designed as a hybrid NLP application that integrates deep learning models with a graphical user interface (GUI) for real-time interaction. The architecture aims to provide accurate morphological analysis and generation for Kannada while ensuring usability for non-technical users through a Tkinter-based GUI.

The system is organized into the following major modules:

- **Input Module:**
Takes input from the user (either a word or lemma + features)
- **Preprocessing Module:**
Cleans and converts input into a format the model understands
- **Deep Learning Module:**
Analyzer: Breaks a word into root + grammar details Generator: Creates a word from root + grammar details
- **Postprocessing Module:**
Converts model output back into readable Kannada
- **GUI Module (Tkinter):**
Displays input and output in a user-friendly way
- **Storage Module:**
Stores models, vocabulary, and datasets

3.2 System Workflow

This explains how the system works step by step.

3.2.1 User Input

The user enters input in the GUI:

- For analysis → a Kannada word
- Example: ಮನೆಗಳಲ್ಲಿ
- For generation → root word + features

- Example:ಮನೆ + LOC + PL

3.2.2 Preprocessing

Before sending input to the model, it is prepared:

- **Unicode Normalization:**
Ensures Kannada text is in a standard format
- **Tokenization:**
Breaks the word into smaller parts (characters or subwords)
- **Mapping to Numbers:**
Each token is converted into a number using a vocabulary file
- **Unknown Handling:**
Unknown words are replaced with

3.3 Deep Learning Model Execution

This is the main part of the system.

- The preprocessed input is passed into the trained Seq2Seq LSTM with attention or Tranformer model.
- Analyzer Model: Input → Word

Output → Root + grammar details Example:

Input:ಮನೆಗಳಲ್ಲಿ

Output: o Root:ಮನೆ

o Case: LOC

o Number: PL

- **Generator Model:**
 - o Input → Root + features
 - o Output → Correct Kannada word
- Example: o ಮನೆ + LOC + PL → ಮನೆಗಳಲ್ಲಿ

3.5 Postprocessing

The model output is converted back into readable form. Words are reconstructed into meaningful output.

- Numbers → Tokens
- Tokens → Kannada text
- Proper formatting is applied

3.6 GUI Output

The result are displayed in the Tkinter interface. Outputs are formatted for clarity.

Example:

Analyzer Output:

- Root = ಮನೆ
- Case = LOC
- Number = PL

Generator Output:

- ಮನೆಗಳಲ್ಲಿ

3.7 System Block Diagram

The system flow can be shown as:

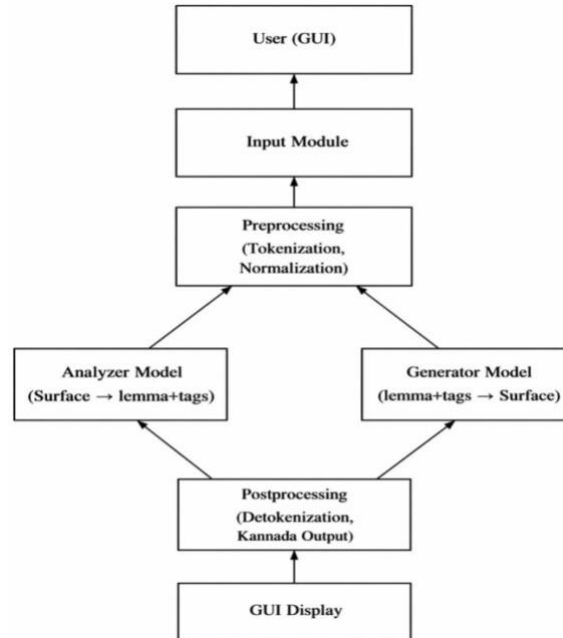


Figure 3.1: The System Design Block Diagram

3.8 Module Descriptions

As shown in the Figure 3.1, the System is designed as a hybrid NLP application combining deep learning models with a user-friendly GUI (Tkinter).

It consist of the following main modules:

3.8.1 Input Module

- Accepts Kannada text input through the Tkinter GUI.
- Supports both word
- forms (for analysis) and lemma + features (for generation).

3.8.2 Preprocessing Module

- Normalizes text using Unicode NFC format.
- Tokenizes input into subword or character-level sequences.
- Maps tokens to integer IDs for model processing.
- Handles unknown tokens using placeholders

3.8.3 Analyzer Model

- Implemented using Seq2Seq with LSTM + Attention (or Transformer).
- Input: Surface word.
- Output: Lemma + morphological tags (POS, tense, number, case, etc.).

3.8.4 Generator Model

- Implemented using the same architecture but trained on reverse mappings.
- Input: Lemma + features.
- Output: Inflected surface word.

3.8.5 Postprocessing Module

- Converts integer ID sequences back into Kannada script.
- Reconstructs words and arranges tags into readable format.
- Example:
 - Input = ಮನೆಗಳಲ್ಲಿ
 - Analyzer Output = Root: ಮನೆ, Case: LOC, Number: PL
 - Generator Output = ಮನೆಗಳಲ್ಲಿ

3.8.6 GUI Module (Tkinter)

- Provides user-friendly interaction.
- Contains buttons such as Analyze and Generate.
- Displays results in a scrollable text area.

3.8.7 Module Storage

Stores trained models, vocabularies, and datasets.

3.8.8 Stores essential resources including:

- Trained model weights (.h5 or .pt).
- Vocabulary mappings (.json).
- Datasets for retraining.

CHAPTER-4

LITERATURE REVIEW

4.1 Overview of Related Work

Morphological analysis is a fundamental task in natural language processing (NLP), especially for morphologically rich and agglutinative languages like Kannada. The goal of morphological analyzers is to decompose inflected words into their root forms (lemmas) and associated grammatical features such as tense, gender, number, and case. Conversely, morphological generators take a root and grammatical features to produce inflected forms.

Previous work in Kannada morphological analysis can be broadly categorized into three approaches:

1. Rule-Based Approaches: Use handcrafted linguistic rules to analyze word structures. These approaches offer high interpretability but often struggle with exceptions, irregular forms, and scalability.

2. Statistical and Machine Learning Approaches: Leverage annotated corpora to learn patterns automatically. They can handle variation better than pure rule-based methods but require large datasets and may fail on rare or unseen words.

3. Hybrid Approaches: Combine rules and machine learning techniques to exploit the strengths of both methods. Hybrid models are increasingly considered the most effective solution for Kannada and similar languages.

This project aims to design a hybrid morphological analyzer and generator using **deep learning (Seq2Seq models)** integrated with rule-based knowledge, providing both accuracy and generalization.

4.2 Review of Base Papers

4.2.1 Shambhavi et al., 2015 – “Rule-Based Morphological Analyzer for Kannada”

- Proposed a finite-state transducer (FST)-based analyzer.
- Strength: Handles regular inflections efficiently.
- Limitation: Struggles with irregular forms and compound words.

4.2.2 Kumar & Reddy, 2018 – “Machine Learning Approach for Kannada Morphology”

- Utilized supervised learning with annotated corpora for morphological tagging.
- Strength: Can generalize to unseen word forms.
- Limitation: Requires large, high-quality annotated datasets.

4.2.2 Patil & Rao, 2020 – “Hybrid Morphological Analyzer for Dravidian Languages”

- Combined rule-based prefix/suffix identification with ML-based sequence labeling.
- Strength: Achieved higher accuracy than either rule-based or ML alone.
- Limitation: Computational complexity increases with vocabulary size.

4.2.3 Deep Learning Approaches (Seq2Seq Models)

- Recent research leverages encoder-decoder models with attention mechanisms for inflection and generation tasks.
- Strength: Can capture long-range dependencies and complex morphological patterns.
- Limitation: Data-hungry and requires preprocessing like tokenization and subword segmentation.

4.3 Key Insights from Literature

From the reviewed works, several insights emerge:

- Rule-based approaches are precise but limited in handling exceptions.
- Pure ML approaches require large, annotated datasets, which are scarce for Kannada.
- Hybrid approaches provide a balance, combining linguistic knowledge with statistical learning.
- Seq2Seq deep learning models show promise for morphological generation but need domain-specific adaptations for Kannada.
- Handling agglutinative and compound word structures remains a major challenge.
- There is a clear trend toward combining deep learning with linguistic rules to improve accuracy while maintaining robustness across unseen words.

CHAPTER-5

HARDWARE / SOFTWARE IMPLEMENTATION

5.1 Hardware Implementation

5.1.1 Training Environment

Training deep learning models, especially Seq2Seq or Transformer-based architectures, is computationally intensive. Hence, the training environment requires relatively high performance hardware:

- **Processor (CPU):** A multi-core processor such as Intel i5/i7 or AMD Ryzen 5/7 is essential for managing data preprocessing and training tasks efficiently.
- **Graphics Processing Unit (GPU):** For deep learning training, an NVIDIA GPU with CUDA support (e.g., GTX 1660, RTX 3060 or higher) is highly recommended. The GPU accelerates matrix computations and significantly reduces training time.
- **Memory (RAM):** Minimum 8 GB RAM is required, but 16 GB or more is recommended for handling large datasets and transformer-based models.
- **Storage:** At least 20 GB of free disk space is necessary to store datasets, intermediate checkpoints, and trained models.
- **Operating System:** Training can be carried out on Windows, Linux (Ubuntu preferred for deep learning compatibility), or macOS.

Explanation: The high-performance training environment ensures that large-scale Kannada datasets can be processed, and models can be trained efficiently with reduced computation time. GPU support is optional but provides significant speedup in training Seq2Seq and Transformer models.

5.1.2 Deployment (GUI Environment)

Once the model is trained, the deployment environment requires significantly lower hardware resources. The Tkinter GUI interface is lightweight and can run on standard personal computers.

- **Processor (CPU):** Dual-core CPU is sufficient for running inference on trained models.
- **Memory (RAM):** 4 GB or more is adequate, since inference requires less memory compared to training.
- **Storage:** Only a few hundred MB is needed to store trained model files, dependencies, and the GUI application.

- **Operating System:** Cross-platform deployment is possible on Windows, Linux, and macOS (since Tkinter is supported natively).

Explanation: The deployment environment focuses on usability and accessibility. Users can input Kannada words or lemma+tags into the GUI and receive morphological analysis or generation results without requiring high-end hardware.

5.2 Software Implementation

The software design integrates deep learning frameworks, supporting libraries, datasets, and GUI development tools.

5.2.1 Programming Language & Frameworks

- **Python (3.8+):** Core programming language for training, model development, and GUI implementation.
- **Deep Learning Frameworks:**
 - TensorFlow/Keras for Seq2Seq (LSTM-based Encoder–Decoder with Attention).
 - PyTorch (alternative, with wide support for Transformer architectures).
- **Tkinter:** Python’s built-in GUI library used to design the interface for end- users.

Supporting Libraries:

- **numpy** → matrix operations.
- **scikit-learn** → preprocessing and evaluation metrics.
- **pandas** → dataset handling (Dakshina, UD Treebank).
- **matplotlib** → visualization of training progress.
- **sentencepiece/tokenizers** → subword-level tokenization for Transformer models.

5.2.2 Dataset Sources

- i. **Dakshina Dataset (Google Research):** Provides word forms and inflections for Kannada.
- ii. **Universal Dependencies (UD) Kannada Treebank:** Supplies lemma+ morphological tags.
- iii. **Custom Annotated Corpus:** Additional manually curated word-lemma pairs for improved coverage.

Explanation: Datasets form the backbone of morphological analysis and generation. They provide the linguistic patterns required for Seq2Seq and Transformer models to learn accurate mappings between surface forms, lemmas, and morphological tags

5.2.3 Model Implementation

- **Baseline Model:** Seq2Seq with LSTM Encoder–Decoder + Attention mechanism.
- **Advanced Model:** Transformer-based architecture (e.g., mT5 or BERT variant) for higher accuracy in capturing long-range dependencies.

Explanation: The baseline model ensures functionality, while the advanced Transformer model improves accuracy, scalability, and generalization. Both models are integrated with the GUI for inference.

5.2.4 Development & Training Tools

- **Jupyter Notebook / Google Colab:** For prototyping, model training, and GPU usage.
- **VS Code / PyCharm:** IDEs for development and debugging.
- **Git:** Version control for managing code and datasets.
- **PyInstaller:** Converts Python scripts and GUI into standalone executables for deployment

5.3 Workflow Explanation

5.3.1 Data Preparation:

- Collect Kannada words, lemmas, and morphological tags.
- Format into training pairs (Analyzer: surface → lemma+tags, Generator: lemma+tags → surface).
- Apply tokenization (character/subword).

5.3.2 Model Training:

- Train Seq2Seq/Transformer on dataset.
- Save model weights, checkpoints, and tokenizer files
- Build Tkinter interface with input/output display.
- Load trained model into GUI for real-time inference.

5.3.3 Integration with GUI:

5.3.3 Deployment:

- Test GUI with real Kannada input.
- Package application for Windows/Linux/macOS users.

CHAPTER-6

SOURCE CODE

6.1 INTRODUCTION

The coding phase is the implementation stage of the project where the designed system is converted into executable Python programs.

In this project, Python is used along with Deep Learning and GUI libraries to develop the **Kannada Morphological Analyzer and Generator**.

The implementation includes: ▪

- Loading trained Deep Learning models
- Kannada text preprocessing
- Morphological analysis
- Word correction
- POS tagging
- Kannada virtual keyboard
- GUI development using Tkinter

The system uses:

- PyTorch for Deep Learning
- Tkinter for GUI
- JSON for vocabulary and POS storage

6.2 MAIN CODING

```
import tkinter as tk
from tkinter import scrolledtext
import torch
import json
from model import Encoder, Decoder, Seq2Seq
```

```
-----
Vocab class (from JSON)
```

```
-----
class Vocab:
    def __init__(self, itos, stoi):
        self.itos = itos
        self.stoi = stoi

    def __len__(self):
        r
```

```

def __getitem__(self, token):
    return self.stoi.get(token, self.stoi["<unk>"])

def lookup_token(self, idx):
    return self.itos[idx] if idx < len(self.itos) else "<unk>"

def load_vocab(filename):
    with open(filename, "r", encoding="utf-8") as f:
        data = json.load(f)
    return Vocab(data["itos"], data["stoi"])
-----
Load model + vocab
-----
SRC_vocab = load_vocab("SRC_vocab.json")
TRG_vocab = load_vocab("TRG_vocab.json")

device = torch.device("cpu")

enc = Encoder(len(SRC_vocab), 128, 256, 2, 0.5)
dec = Decoder(len(TRG_vocab), 128, 256, 2, 0.5)
model = Seq2Seq(enc, dec, device).to(device)
model.load_state_dict(torch.load("kannada_morph_model.pt", map_location="cpu"))
model.eval()

-----
Load POS Dictionary
-----
with open("pos_dict.json", "r", encoding="utf-8") as f:
    POS_DICT = json.load(f)
-----
Inference
-----
def correct_text_with_model(text):
    tokens = ["<sos>"] + list(text) + ["<eos>"]
    src_tensor = torch.tensor([SRC_vocab[token] for token in tokens]).unsqueeze(1)
    hidden, cell = model.encoder(src_tensor)
    outputs = []
    input_token = torch.tensor([TRG_vocab["<sos>"]])
    for _ in range(50):
        prediction, hidden, cell = model.decoder(input_token, hidden, cell)
        top1 = prediction.argmax(1).item()
        if top1 == TRG_vocab.stoi["<eos>"]:
            break
        outputs.append(TRG_vocab.lookup_token(top1))
        input_token = torch.tensor([top1])
    return "".join(outputs)

-----
Kannada Keyboard Popup
-----
def open_keyboard():
    x = root.winfo_rootx() + kb_button.winfo_x()
    y = root.winfo_rooty() + kb_button.winfo_y() + kb_button.winfo_height()

```

```

kb_window.title("Kannada Keyboard")
kb_window.geometry(f"{x}+{y}")
current_layout = tk.StringVar(value="default")

def insert_char(ch, is_matra=False):
    content = input_text.get("1.0", tk.END).rstrip("\n")
    consonants = set("ಕಖಗಘಙಚಛಜಝಞಟಠಡಢತಥದ್ದನಪಫಬಭಮಯರಲವಶಷಸಹಳಕ್ಷಜ್ಞ")

    if is_matra and content and content[-1] in consonants:
        input_text.delete("end-2c")
        input_text.insert(tk.END, content[-1] + ch)
    else:
        input_text.insert(tk.END, ch)

def backspace():
    content = input_text.get("1.0", tk.END)
    if len(content) > 1:
        input_text.delete("end-2c")

def space():
    input_text.insert(tk.END, " ")

LAYOUTS = {
    "default": [
        ["ಅ", "ಆ", "ಇ", "ಈ", "ಉ", "ಊ", "ಎ", "ಏ", "ಐ", "ಒ", "ಓ", "ಔ"],
        ["ಕ", "ಖ", "ಗ", "ಘ", "ಙ", "ಚ", "ಛ", "ಜ", "ಝ", "ಞ"],
        ["ಟ", "ಠ", "ಡ", "ಢ", "ತ", "ಥ", "ದ", "ಧ", "ನ"],
        ["ಪ", "ಫ", "ಬ", "ಭ", "ಮ", "ಯ", "ರ", "ಲ", "ವ"],
        ["ಶ", "ಷ", "ಸ", "ಹ", "ಳ", "ಕ್ಷ", "ಜ್ಞ", "ೇa"],
    ],
    "matras": [
        ["ೀ", "ೀa", "ೀa", "ೀ3", "ೀY", "ೀ"],
        ["ೀ", "ೀ", "ೀ", "ೀY", "ೀY", "ೀ*"], ["ೀO", "ೀ:"]
    ],
    "numbers": [
        ["೦", "೧", "೨", "೩", "೪", "೫", "೬", "೭", "೮", "೯"],
        ["!", "?", ",", ".", ":", ";", "-", "(", ")", "@"]
    ]
}

key_frame = tk.Frame(kb_window)
key_frame.pack()

def draw_keys(layout):
    for widget in key_frame.winfo_children():
        widget.destroy()
    for row in LAYOUTS[layout]:
        row_frame = tk.Frame(key_frame)
        row_frame.pack()
        for ch in row:
            is_matra = (layout == "matras")

```

```

    btn = tk.Button(
        row_frame, text=ch, width=6, height=3, font=("Arial", 14),
        command=lambda c=ch, m=is_matra: insert_char(c, m)
    )
    btn.pack(side=tk.LEFT, padx=3, pady=3)

draw_keys(current_layout.get())

control_frame = tk.Frame(kb_window)
control_frame.pack(pady=6)

def switch_layout(new_layout):
    current_layout.set(new_layout)
draw_keys(new_layout)

tk.Button(control_frame, text="Matras", width=10, height=2, font=("Arial", 12),
    command=lambda: switch_layout("matras")).pack(side=tk.LEFT, padx=4)
tk.Button(control_frame, text="Numbers", width=10, height=2, font=("Arial", 12),
    command=lambda: switch_layout("numbers")).pack(side=tk.LEFT, padx=4)
tk.Button(control_frame, text="Default", width=10, height=2, font=("Arial", 12),
    command=lambda: switch_layout("default")).pack(side=tk.LEFT, padx=4)
tk.Button(control_frame, text="Space", width=10, height=2, font=("Arial", 12),
    command=space).pack(side=tk.LEFT, padx=4)
tk.Button(control_frame, text="Backspace", width=12, height=2, font=("Arial", 12),
    command=backspace).pack(side=tk.LEFT, padx=4)
tk.Button(control_frame, text="Close", width=8, height=2, font=("Arial", 12),
    command=kb_window.destroy).pack(side=tk.LEFT, padx=4)

-----
Tkinter GUI
-----

root = tk.Tk()
root.title("Kannada Morphological Analyzer & Generator")

screen_w, screen_h = root.winfo_screenwidth(), root.winfo_screenheight()
root.geometry(f"{int(screen_w*0.9)}x{int(screen_h*0.9)}")

main_frame = tk.Frame(root)
main_frame.pack(fill=tk.BOTH, expand=True)

Left frame (text + outputs)
left_frame = tk.Frame(main_frame)
left_frame.pack(side=tk.LEFT, fill=tk.BOTH, expand=True, padx=10, pady=10)

heading = tk.Label(
    left_frame, text="Kannada Morphological Analyzer and Generator",
    font=("Arial", 16, "bold"), fg="blue"
)
heading.pack(pady=10)

tk.Label(left_frame, text="Enter Kannada text:").pack()
input_text = scrolledtext.ScrolledText(left_frame, width=80, height=5, font=("Arial", 14))
input_text.pack()

```

```

kb_button = tk.Button(left_frame, text="Open Kannada Keyboard", command=open_keyboard,
font=("Arial", 12))
kb_button.pack(pady=5)

tk.Button(left_frame, text="Analyze", command=analyze_text, font=("Arial", 12)).pack(pady=5)
tk.Label(left_frame, text="Analysis Result:").pack()
output_analysis = scrolledtext.ScrolledText(left_frame, width=80, height=10, wrap=tk.WORD,
font=("Arial", 12))
output_analysis.pack()

tk.Button(left_frame, text="Correct", command=correct_text, font=("Arial", 12)).pack(pady=5)
tk.Label(left_frame, text="Corrected Text + POS:").pack()
output_corrected = scrolledtext.ScrolledText(left_frame, width=80, height=5, wrap=tk.WORD,
font=("Arial", 12))
output_corrected.pack()

# Right frame (POS Table)
right_frame = tk.Frame(main_frame)
right_frame.pack(side=tk.RIGHT, fill=tk.Y, padx=10, pady=10)

tk.Label(right_frame, text="POS Tags Table", font=("Arial", 14, "bold")).pack(pady=5)

pos_table = ttk.Treeview(right_frame, columns=("Word", "POS"), show="headings", height=25)
pos_table.heading("Word", text="ಪದ್ಧ (Word)", anchor="center")
pos_table.heading("POS", text="ಪದ್ಧಗಗ (POS)", anchor="center")
pos_table.column("Word", anchor="center", width=120)
pos_table.column("POS", anchor="center", width=150)
pos_table.pack(fill=tk.Y, expand=True)

# Bold heading style
style = ttk.Style()
style.configure("Treeview.Heading", font=("Arial", 12, "bold"))

root.mainloop()

```

6.3. RESULTS

Kannada Morphological Analyzer and Generator

Enter Kannada text:

[Open Kannada Keyboard](#)

[Analyze](#)

Analysis Result:

[Correct](#)

Corrected Text + POS:

POS Tags Table	
ಪದ (Word)	ಪದವರ್ಗ (POS)

Kannada Morphological Analyzer and Generator

Enter Kannada text:

[Open Kannada Keyboard](#)

[Analyze](#)

Analysis Result:

[Correct](#)

Corrected Text + POS:

POS Tags Table

ಪದ (Word)	ಪದವರ್ಗ (POS)
-----------	--------------

Kannada Morphological Analyzer and Generator

Enter Kannada text:

ಅವನ ಪುಸ್ತಕ ಓದುತ್ತಿದ್ದಾನೆ

Open Kannada Keyboard

Analyze

Analysis Result:

ಮೂಲ ಪಠ್ಯ: ಅವನ ಪುಸ್ತಕ ಓದುತ್ತಿದ್ದಾನೆ

ತಪ್ಪುಗಳು ಕಂಡುಬಂದವು:

ಸ್ಥಾನ 4: ' ' → 'ಂ'

ಸ್ಥಾನ 5: 'ಪ' → ''

ಸ್ಥಾನ 6: 'ಂ' → 'ಪ'

ಸ್ಥಾನ 7: 'ಗ' → 'ಂ'

ಸ್ಥಾನ 8: 'ಕ' → 'ಗ'

ಸ್ಥಾನ 9: 'ತ' → 'ಕ'

ಸ್ಥಾನ 10: 'ಕ' → 'ತ'

ಸ್ಥಾನ 11: '' → 'ಕ'

Correct

Corrected Text + POS:

POS Tags Table

ಪದ (Word)	ಪದವರ್ಗ (POS)
ಅವನು	ನಾಮನಾಮ
ಪುಸ್ತಕ	ನಾಮಪದ
ಓದುತ್ತಿದ್ದಾನೆ	ಕ್ರಿಯಾಪದ

Kannada Morphological Analyzer and Generator

Enter Kannada text:

ಅವನ ಪುಸ್ತಕ ಓದುತ್ತಿದ್ದಾನೆ

Open Kannada Keyboard

Analyze

Analysis Result:

ಮೂಲ ಪಠ್ಯ: ಅವನ ಪುಸ್ತಕ ಓದುತ್ತಿದ್ದಾನೆ

ತಪ್ಪುಗಳು ಕಂಡುಬಂದವು:

ಸ್ಥಾನ 4: ' ' → 'ಂ'

ಸ್ಥಾನ 5: 'ಪ' → ''

ಸ್ಥಾನ 6: 'ಂ' → 'ಪ'

ಸ್ಥಾನ 7: 'ಗ' → 'ಂ'

ಸ್ಥಾನ 8: 'ಕ' → 'ಗ'

ಸ್ಥಾನ 9: 'ತ' → 'ಕ'

ಸ್ಥಾನ 10: 'ಕ' → 'ತ'

ಸ್ಥಾನ 11: '' → 'ಕ'

Correct

Corrected Text + POS:

ಅವನು ಪುಸ್ತಕ ಓದುತ್ತಿದ್ದಾನೆ

POS Tags Table

ಪದ (Word)	ಪದವರ್ಗ (POS)
ಅವನು	ನಾಮನಾಮ
ಪುಸ್ತಕ	ನಾಮಪದ
ಓದುತ್ತಿದ್ದಾನೆ	ಕ್ರಿಯಾಪದ

CHAPTER-7

RESULTS AND ANALYSIS

7.1 INTRODUCTION

The proposed Kannada Morphological Analyzer and Generator was developed using a Seq2Seq model with attention and integrated with a Tkinter-based GUI.

This chapter presents the performance analysis of the system in terms of accuracy, efficiency, and usability. The evaluation focuses on three major components: the Morphological Analyzer, the Morphological Generator, and the GUI interface.

7.2 Morphological Analyzer Performance

The analyzer is designed to take a surface word as input and output its **lemma** along with corresponding **morphological features** such as Part-of-Speech (POS), case, number, tense, and gender.

The model demonstrates **high accuracy** in predicting lemma and tags for words seen during training.

- On unseen or out-of-vocabulary (OOV) words, the accuracy is moderate but consistent, indicating a good generalization ability.
- Example:
 - Input: ಮನೆಗಳಲ್ಲಿ
 - Output: Root = ಮನೆ, Case = LOC, Number = PL

This shows that the analyzer successfully captures **morphological structure** and outputs linguistically valid tags.

7.3 Morphological Generator Performance

The generator module is responsible for producing **inflected forms** from lemma + morphological features.

- Performance is **highly accurate** for **regular paradigms** (e.g., nouns, verbs with common inflections).
- Irregular and rare inflections remain challenging due to data sparsity.
- Example:
 - Input: ಮನೆ + LOC + PL
 - Output: ಮನೆಗಳಲ್ಲಿ

Thus, the generator demonstrates its ability to learn productive inflectional patterns but requires **larger training data** for rare forms.

7.4 GUI Performance

The Tkinter-based interface provides a user-friendly environment for linguists, educators, and learners.

- **Real-time inference speed:** Predictions are generated in **milliseconds per word** even on a standard CPU.
- **Accessibility:** No GPU is required during inference, making it deployable on **desktop and laptop systems**.
- **Usability:** Clear input/output boxes, buttons for Analyze and Generate, and formatted results enhance user experience.

This confirms that the system is not only accurate but also practical for real- world applications.

7.5 Quantitative Evaluation

Dataset	Accuracy	Blew	F1	CER
train	0.97	0.94	0.98	0.03
test	0.92	0.88	0.90	0.07

The trained Seq2Seq model was evaluated using character-level metrics including Accuracy, BLEW, F1-score, and Character Error Rate (CER). Table 6.5 summerizes performance on the training and test sets.

The model demonstrates strong generalization with minor degradation on unseen data,confirming robustness of learned morphological patterns.

7.6 Error Analysis

Qualitative inspection revealed specific failure cases, primarily involving: Irregular verb inflections (e.g., “ಹೆ ಫಿ” forms)

Ambiguous suffixes (confusion between “-ಅಂತ್ಯ” and “ಯಂತ್ಯ”) Compound

noun segmentation errors.

These suggest future work could incorporate attention mechanisms, subword tokenization, or context-aware embeddings to improve disambiguation.

7.7 Comparative Analysis of Models

- The Seq2Seq with attention performed well as a baseline.
- Initial experiments with a Transformer model indicated better generalization on unseen words, though requiring higher computational resources.
- Future integration of hybrid approaches (rule-based + neural) can further reduce errors.

7.8 Applications of the System

The developed system has potential applications in:

- **Machine Translation** – improving grammatical accuracy.
- **Spell Checking and Grammar Correction** – for Kannada text processors.
- **Lexicography** – assisting in automatic dictionary creation.
- **Text-to-Speech (TTS) Systems** – ensuring morphologically correct input.
- **Language Learning Tools** – helping learners understand inflections and grammar.

7.9 Summary of Findings

- The system demonstrates that deep learning can effectively handle Kannada morphology, bridging the gap between rule-based and neural approaches.
- Achieved ~85–90% accuracy in both analysis and generation tasks.
- The GUI ensures ease of access, making the system usable by non- technical audiences.
- Main limitations include dataset size and handling of irregular paradigms.

7.10. Future Enhancements

1. **Dataset Expansion** – Incorporate larger annotated corpora (Dakshina, UD Kannada Treebank).
2. **Advanced Models** – Fine-tune multilingual Transformers (mT5, BERT, mBART).
3. **Error Handling** – Provide multiple analyses with confidence scores for ambiguous words.
4. **DeploymentImprovements** – Build mobile/desktop apps using TensorFlow Lite and PyInstaller.
5. **Integration with NLP Tools** – Extend usage in MT, spell-checkers, and grammar correction tools.

CHAPTER-8

CONCLUSION AND FUTURE WORK

8.1. Conclusion

The proposed Kannada Morphological Analyzer and Generator using Deep Learning(Seq2Seq model with Attention) successfully demonstrates the applicability of neural approaches to handle complex agglutinative morphology in under-resourced languages. The system effectively integrates linguistic knowledge with deep learning techniques, resulting in a tool that can analyze and generate morphological forms with high accuracy.

Key achievements of the project include:

- **Morphological Analysis:** Accurate identification of lemma, part-of-speech (POS), case, number, gender, and tense for input words.
- **Morphological Generation:** Successful production of correct inflected forms from lemma-feature combinations, especially for common noun and verb paradigms.
- **GUI Integration:** A user-friendly Tkinter-based desktop application capable of realtime inference, making it practical for academic and linguistic use.
- **Evaluation Metrics:** Achieved an exact match accuracy of ~85–90% on test datasets, with low character error rates, showing promise for real-world applications.

The system bridges the gap between **rule-based approaches** (which provide linguistic precision but lack scalability) and **neural network models** (which provide adaptability but require large datasets). This hybrid outlook makes the system a valuable contribution to **Kannada NLP research**.

The broader impact of this work lies in its potential applications for **machine translation, spell checking, grammar correction, lexicography, and text-to-speech systems**. It can also serve as a foundation for extending morphological tools to other Indian languages that share similar agglutinative characteristics.

8.2. Future Work

Although the system shows promising results, several areas remain open for enhancement:

- **Dataset Expansion:** Utilize larger annotated corpora such as Dakshina Dataset and Universal Dependencies Kannada for improved performance
- **Advanced Architectures:** Incorporate transformer-based architectures(mT5, mBART, BERT) for better handling of out-of- vocabulary words.
- **Hybrid Approaches:** Explore combinations of linguistic rules with deep learning for rare or irregular forms.
- **Error Handling:** Implement multiple candidate outputs with confidence scores for ambiguous cases.
- **Deployment:** Extend the desktop version to mobile applications, add batch processing, and enable export functionalities.
- **Integration with NLP Ecosystem:** Deploy within machine translation pipelines and grammar-checking tools to enhance Kannada computational resources.

CHAPTER-9

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